In [ ]:
```
# EXPERIMENT 8 – DATA PRE-PROCESSING & FEATURE ENGINEERING
PIN NO.: 241UDS0099
## AIM
To implement data pre-processing and feature engineering techniques on a dataset, including removing unnecessary features, feature
scaling, and encoding categorical variables using Python libraries.
```

In [ ]:
```
## Step 1: Import the Basic Libraries
```

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
## Step 2: Load & Analyze the Dataset
```

In [ ]:
```
df_241UDS0099 = pd.read_csv("Experiment_8_241UDS0099_Academic_Feature_Preprocessing.csv")
df_241UDS0099.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 121 entries, 0 to 120
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Branch                 118 non-null    str
 1   Marks[10th]            118 non-null    float64
 2   Marks[12th]            117 non-null    float64
 3   Gender                 119 non-null    str
 4   Board[10th]            119 non-null    str
 5   Board[12th]            119 non-null    str
 6   Category               119 non-null    str
 7   GPA 1                  119 non-null    float64
 8   Rank                   120 non-null    float64
 9   Normalized Rank        119 non-null    float64
 10  CGPA                   117 non-null    float64
 11  Current Back           119 non-null    float64
 12  Ever Back              118 non-null    float64
 13  GPA 2                  119 non-null    float64
 14  GPA 3                  119 non-null    float64
 15  GPA 4                  119 non-null    float64
 16  GPA 5                  119 non-null    float64
 17  GPA 6                  119 non-null    float64
 18  Olympiads Qualified    119 non-null    float64
 19  Technical Projects     119 non-null    float64
 20  Tech Quiz              119 non-null    float64
 21  Engg. Coaching         119 non-null    float64
 22  NTSE Scholarships      119 non-null    float64
 23  Miscellany Tech Events  119 non-null    float64
dtypes: float64(19), str(5)
memory usage: 22.8 KB
```

In [ ]:
```
df_241UDS0099.head()
```

Out[4]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	75.50	Female	State Board	BIEAP	SC
1	EEE	81.08	75.03	Male	State Board	CBSE	ST
2	CSE	76.76	77.64	Male	ICSE	State Board	EWS
3	MECH	83.50	77.14	Female	BSEB Patna	ISC	OBC
4	ECE	90.24	85.93	Female	CBSE	ISC	ST

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	6.59	38129.0	0.635483	...	6.99	6.98	6.29
1	6.51	55894.0	0.931567	...	6.36	6.76	5.85
2	7.05	50405.0	0.840083	...	6.98	7.21	6.72
3	7.55	32040.0	0.534000	...	7.41	7.86	8.06
4	7.42	25211.0	0.420183	...	7.27	7.37	7.69

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	6.43	2.0	1.0	2.0	2.0	0.0	3.0
1	6.60	0.0	2.0	4.0	2.0	1.0	4.0
2	7.04	0.0	4.0	1.0	2.0	1.0	1.0
3	7.75	0.0	1.0	0.0	0.0	1.0	0.0
4	6.57	0.0	0.0	0.0	1.0	0.0	2.0

5 rows × 24 columns

In [ ]:
```
df_241UDS0099.shape
```

Out[5]:
```
(121, 24)
```

In [ ]:
```
df_241UDS0099.isnull().sum()
```

Out[6]:
```
Branch                   3
Marks[10th]              3
Marks[12th]              4
Gender                   2
Board[10th]              2
Board[12th]              2
Category                 2
GPA 1                    2
Rank                     1
Normalized Rank          2
CGPA                     4
Current Back             2
Ever Back                3
GPA 2                    2
GPA 3                    2
GPA 4                    2
GPA 5                    2
GPA 6                    2
Olympiads Qualified      2
Technical Projects       2
Tech Quiz                2
Engg. Coaching           2
NTSE Scholarships        2
Miscellany Tech Events   2
dtype: int64
```

In [ ]:
```
df_241UDS0099 = df_241UDS0099.dropna(how="all")
df_241UDS0099.shape
```

Out[7]:
```
(120, 24)
```

In [ ]:
```
## Step 3: Remove Unnecessary Features
```

In [ ]:
```
df_241UDS0099 = df_241UDS0099.drop("Rank", axis=1)
df_241UDS0099.shape
```

Out[8]:
```
(120, 23)
```

In [ ]:
```
df_241UDS0099.columns
```

Out[9]:
```
Index(['Branch', 'Marks[10th]', 'Marks[12th]', 'Gender', 'Board[10th]',
       'Board[12th]', 'Category', 'GPA 1', 'Normalized Rank', 'CGPA',
       'Current Back', 'Ever Back', 'GPA 2', 'GPA 3', 'GPA 4', 'GPA 5',
       'GPA 6', 'Olympiads Qualified', 'Technical Projects', 'Tech Quiz',
       'Engg. Coaching', 'NTSE Scholarships', 'Miscellany Tech Events'],
      dtype='str')
```

In [ ]:
```
## Step 4: Scale the Numerical Features
```

In [10]:
```
from sklearn.preprocessing import StandardScaler
```

In [ ]:
```
z_scaler_241UDS0099 = StandardScaler()
```

In [ ]:
```
df_241UDS0099[["Marks[10th]", "Marks[12th]"]] = z_scaler_241UDS0099.fit_transform(
    df_241UDS0099[["Marks[10th]", "Marks[12th]"]]
)
df_241UDS0099.head(3)
```

Out[12]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	-0.235751	Female	State Board	BIEAP	SC
1	EEE	-0.141916	-0.283243	Male	State Board	CBSE	ST
2	CSE	-0.678547	-0.019510	Male	ICSE	State Board	EWS

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	6.59	0.635483	8.85	...	6.99	6.98	6.29
1	6.51	0.931567	7.76	...	6.36	6.76	5.85
2	7.05	0.840083	NaN	...	6.98	7.21	6.72

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	6.43	2.0	1.0	2.0	2.0	0.0	3.0
1	6.60	0.0	2.0	4.0	2.0	1.0	4.0
2	7.04	0.0	4.0	1.0	2.0	1.0	1.0

3 rows × 23 columns

In [17]:
```
from sklearn.preprocessing import MinMaxScaler
```

In [ ]:
```
mm_scaler_241UDS0099 = MinMaxScaler()
```

In [ ]:
```
df_241UDS0099[["CGPA", "GPA 1"]] = mm_scaler_241UDS0099.fit_transform(
    df_241UDS0099[["CGPA", "GPA 1"]]
)
df_241UDS0099.head(3)
```

Out[19]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	NaN	-0.235751	Female	State Board	BIEAP	SC
1	EEE	-0.141916	-0.283243	Male	State Board	CBSE	ST
2	CSE	-0.678547	-0.019510	Male	ICSE	State Board	EWS

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	0.419134	0.635483	0.945238	...	6.99	6.98	6.29
1	0.400911	0.931567	0.685714	...	6.36	6.76	5.85
2	0.523918	0.840083	NaN	...	6.98	7.21	6.72

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	6.43	2.0	1.0	2.0	2.0	0.0	3.0
1	6.60	0.0	2.0	4.0	2.0	1.0	4.0
2	7.04	0.0	4.0	1.0	2.0	1.0	1.0

3 rows × 23 columns

In [ ]:
```
## Step 5: Encode the Categorical Features
```

In [20]:
```
from sklearn.preprocessing import LabelEncoder
```

In [ ]:
```
encoder_241UDS0099 = LabelEncoder()
```

In [ ]:
```
for col in ["Gender", "Category", "Branch"]:
    df_241UDS0099[col] = encoder_241UDS0099.fit_transform(df_241UDS0099[col].astype(str))
df_241UDS0099.head(3)
```

Out[22]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	7	NaN	-0.235751	0	State Board	BIEAP	3
1	3	-0.141916	-0.283243	1	State Board	CBSE	4
2	1	-0.678547	-0.019510	1	ICSE	State Board	0

Columns 9–15 of 22 · first column repeated

	GPA 1	Normalized Rank	CGPA	...	GPA 3	GPA 4	GPA 5
0	0.419134	0.635483	0.945238	...	6.99	6.98	6.29
1	0.400911	0.931567	0.685714	...	6.36	6.76	5.85
2	0.523918	0.840083	NaN	...	6.98	7.21	6.72

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	6.43	2.0	1.0	2.0	2.0	0.0	3.0
1	6.60	0.0	2.0	4.0	2.0	1.0	4.0
2	7.04	0.0	4.0	1.0	2.0	1.0	1.0

3 rows × 23 columns

In [ ]:
```
df_241UDS0099 = pd.get_dummies(
    df_241UDS0099,
    columns=["Board[10th]", "Board[12th]"],
    drop_first=True
)
df_241UDS0099.head(3)
```

Out[23]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Category	GPA 1	Normalized Rank
0	7	NaN	-0.235751	0	3	0.419134	0.635483
1	3	-0.141916	-0.283243	1	4	0.400911	0.931567
2	1	-0.678547	-0.019510	1	0	0.523918	0.840083

Columns 9–15 of 22 · first column repeated

	CGPA	Current Back	Ever Back	...	Tech Quiz	Engg. Coaching	NTSE Scholarships
0	0.945238	0.0	2.0	...	2.0	2.0	0.0
1	0.685714	2.0	1.0	...	4.0	2.0	1.0
2	NaN	0.0	0.0	...	1.0	2.0	1.0

Columns 16–22 of 22 · first column repeated

	Miscellany Tech Events	Board[10th]_CBSE	Board[10th]_ICSE	Board[10th]_State Board	Board[12th]_CBSE	Board[12th]_ISC	Board[12th]_State Board
0	3.0	False	False	True	False	False	False
1	4.0	False	False	True	True	False	False
2	1.0	False	True	False	False	False	True

3 rows × 27 columns

In [ ]:
```
bool_columns_241UDS0099 = df_241UDS0099.select_dtypes(include="bool").columns
for col in bool_columns_241UDS0099:
    df_241UDS0099[col] = df_241UDS0099[col].astype(int)
df_241UDS0099.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 27 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Branch                   120 non-null    int64
 1   Marks[10th]              118 non-null    float64
 2   Marks[12th]              117 non-null    float64
 3   Gender                   120 non-null    int64
 4   Category                 120 non-null    int64
 5   GPA 1                    119 non-null    float64
 6   Normalized Rank          119 non-null    float64
 7   CGPA                     117 non-null    float64
 8   Current Back             119 non-null    float64
 9   Ever Back                118 non-null    float64
 10  GPA 2                    119 non-null    float64
 11  GPA 3                    119 non-null    float64
 12  GPA 4                    119 non-null    float64
 13  GPA 5                    119 non-null    float64
 14  GPA 6                    119 non-null    float64
 15  Olympiads Qualified      119 non-null    float64
 16  Technical Projects       119 non-null    float64
 17  Tech Quiz                119 non-null    float64
 18  Engg. Coaching           119 non-null    float64
 19  NTSE Scholarships        119 non-null    float64
 20  Miscellany Tech Events   119 non-null    float64
 21  Board[10th]_CBSE         120 non-null    int64
 22  Board[10th]_ICSE         120 non-null    int64
 23  Board[10th]_State Board  120 non-null    int64
 24  Board[12th]_CBSE         120 non-null    int64
 25  Board[12th]_ISC          120 non-null    int64
 26  Board[12th]_State Board  120 non-null    int64
dtypes: float64(18), int64(9)
memory usage: 25.4 KB
```

In [ ]:
```
df_241UDS0099.to_csv("Experiment_8_241UDS0099_Processed_Dataset.csv", index=False)
print("Processed dataset saved as: Experiment_8_241UDS0099_Processed_Dataset.csv")
```

```
Processed dataset saved as: Experiment_8_241UDS0110_Processed_Dataset.csv
```

In [ ]:
```
## RESULT
The dataset was preprocessed by removing an unnecessary feature, scaling selected numerical features with StandardScaler and
MinMaxScaler, encoding categorical features using LabelEncoder and one-hot encoding, and converting Boolean columns to integers.
```